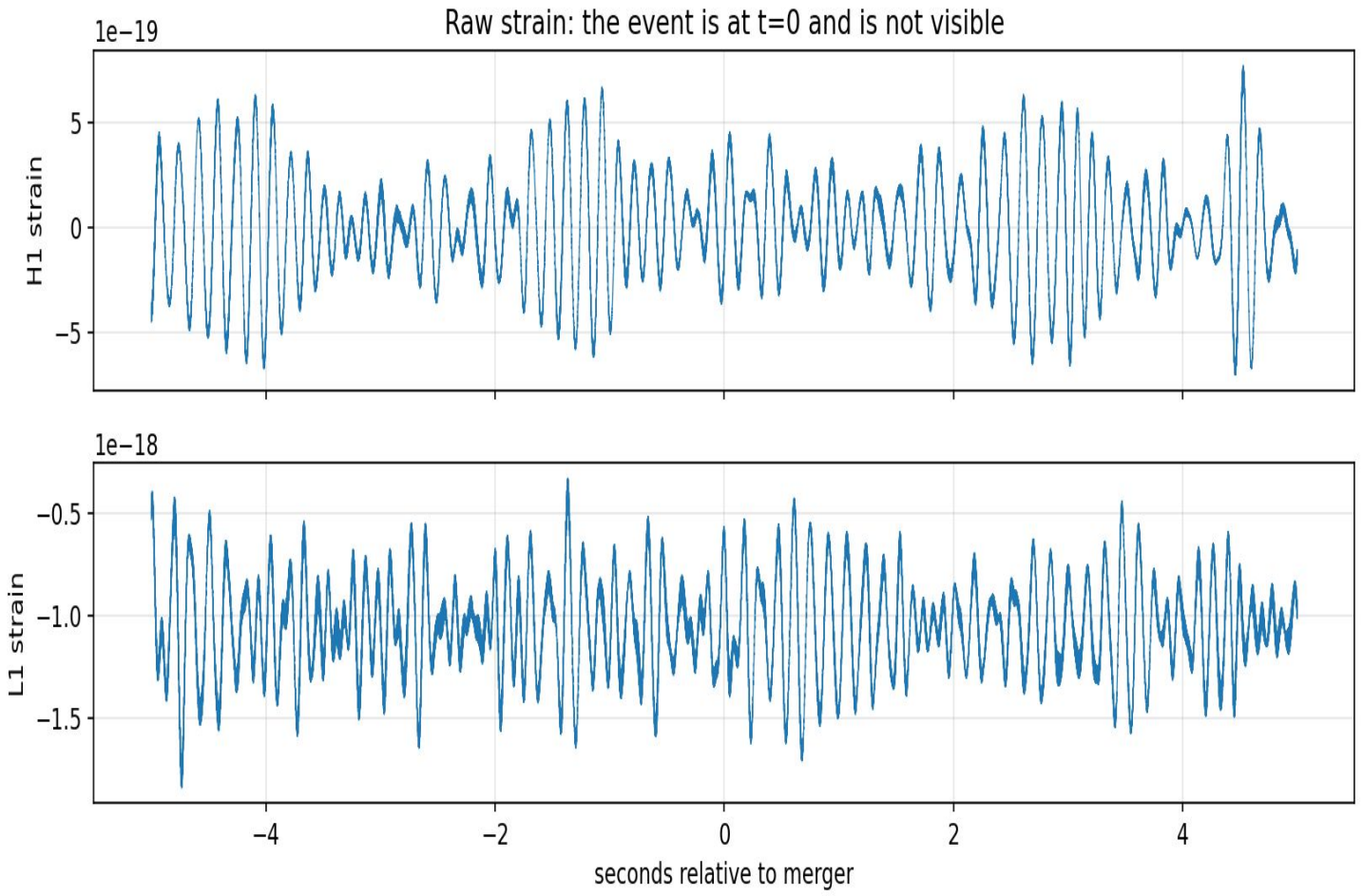
Detecting a binary black hole merger

Andreh Yaghoubian

Raw Strain - the signal is invisible

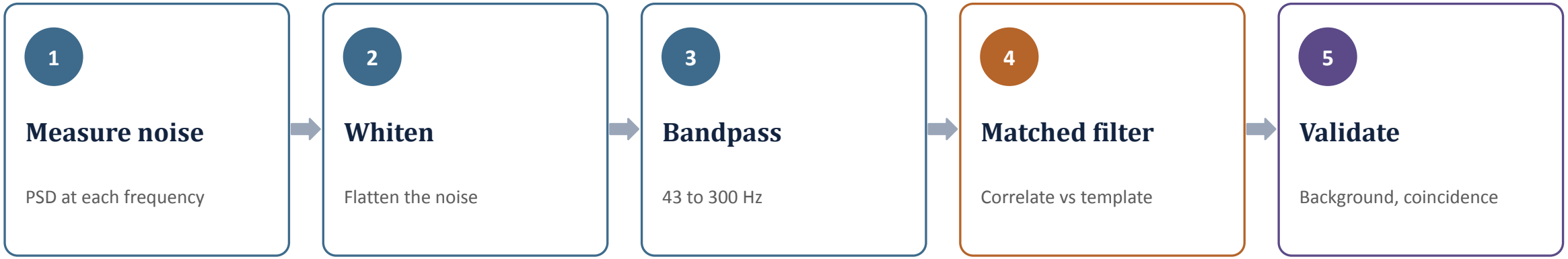
Raw strain from both detectors

- LIGO measures strain: the fractional change in the length of two 4 km arms
- This event produces a strain near 10^{-19}, a length change of about 4 x 10^{-16} m over 4 km, smaller than the width of a proton.
- Detector noise exceeds the signal, therefore nothing is visible in the raw data.



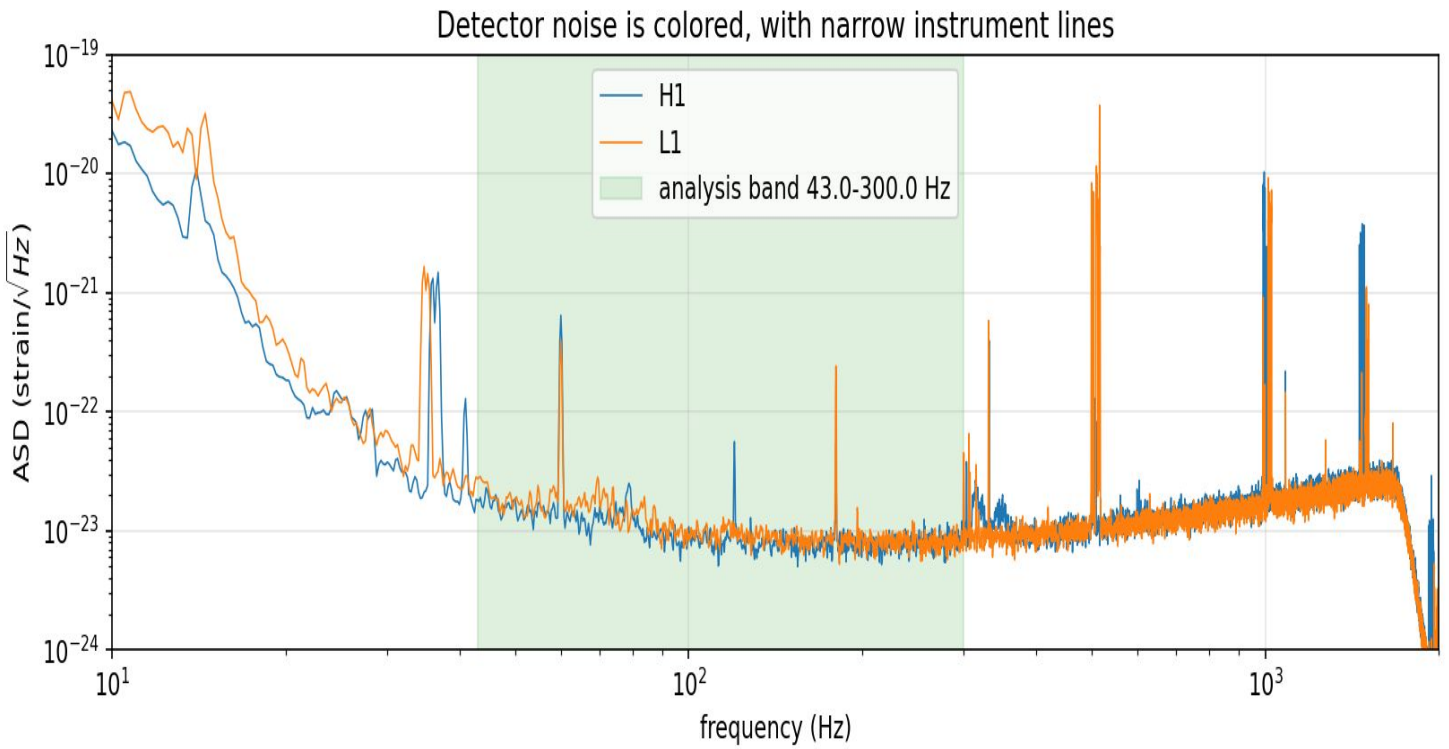
Raw strain, H1 (top) and L1 (bottom). The merger is at t = 0.

Five-stage detection pipeline



Step 1: Measure the noise

Welch periodogram



You cannot correct for noise until you've measured it.

- Colored noise: ~ 1000x larger at 20 Hz than at 100 Hz
- Low frequency - seismic
  high frequency - quantum noise
- Narrow spikes - instrument lines
  - 60 Hz power harmonics, mirror resonances
- Shaded band, 43 to 300 Hz, is where the detector is quiet and the signal has power

Step 2: Whitening (and a bug)

Divide each frequency by its own noise amplitude

0.9979

Synthetic white noise, expect 1.0

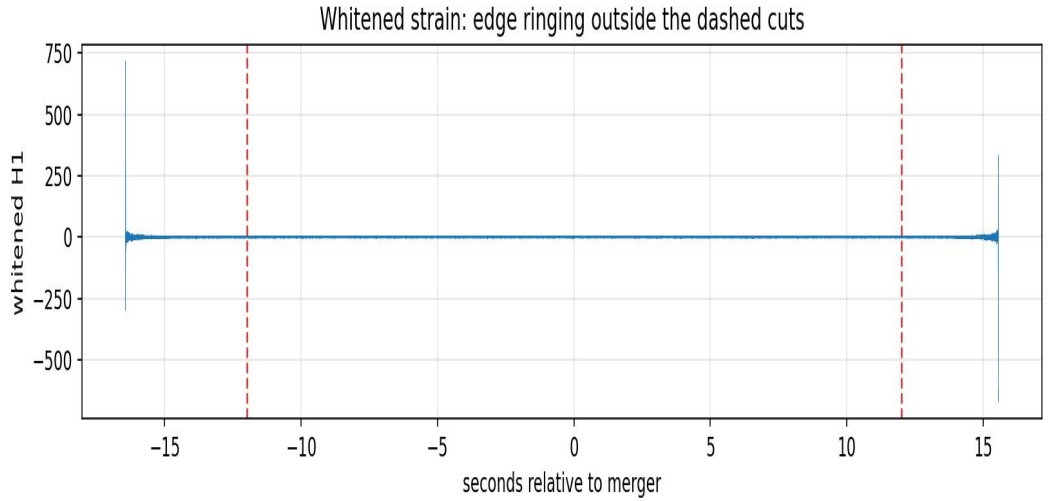
6.273

Real data, first attempt

0.995

After trimming 4 s per end

- Problem: Whitened variance approximately six times higher than expected
- Controlled test: ran the whitener on synthetic noise I generated, where the answer had to be 1.0. It returned 0.998, which cleared the code
- Diagnosis: the FFT treats the 32 s record as periodic, therefore the mismatch between the last and first sample looks like a step. A step has energy at every frequency, including the quiet ones that whitening amplifies most
- Fix: trim 4 s from each end. The event is centered, therefore nothing is lost
- Better fix: taper the ends to zero with a window function, removing the discontinuity instead of discarding data

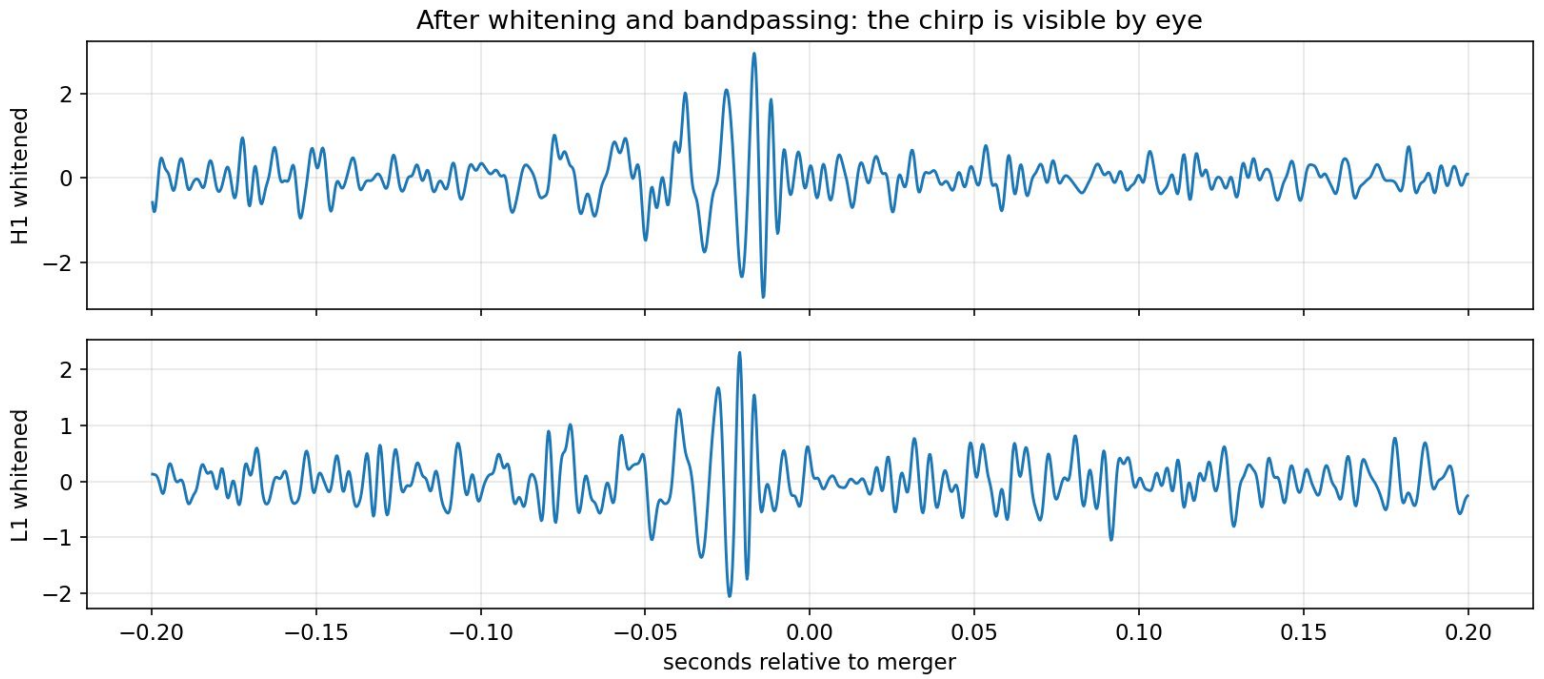


Edge ringing sits outside the dashed cuts

Step 3: Bandpass, and the chirp appears

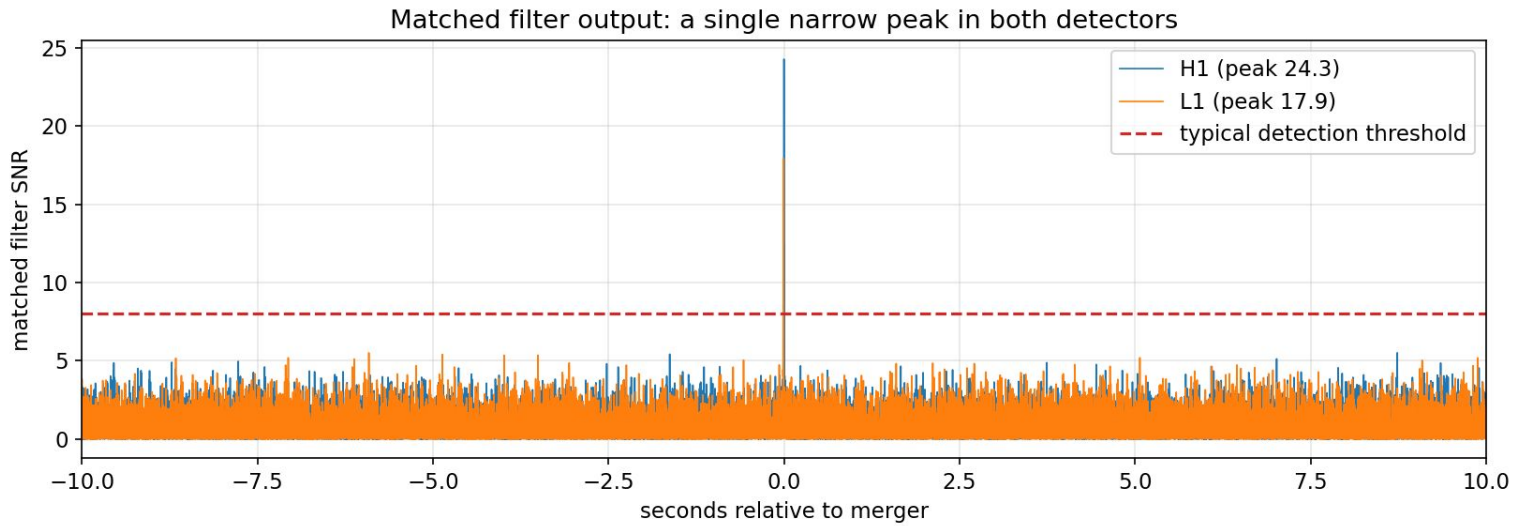
Same data as slide 2

- Keep 43 to 300 Hz, discard the rest
- Wiggles start wide and slow, then get faster and tighter as amplitude grows. That is frequency sweeping upward: the chirp
- t = 0 is the merger, therefore the visible burst is the inspiral leading up to it
- Zero-phase filtering avoids shifting the waveform in time, which matters because arrival time is a measurement



Step 4: Matched filter

Correlate against a known waveform, weighted by inverse noise



24.25

H1 peak SNR

17.92

L1 peak SNR

0.4 ms

From the published merger time

- Output is signal-to-noise ratio as a function of arrival time. The peak says both when the signal arrived and how strong it was relative to noise
- The filter never receives the known event time. It searches all 32 s and reports the maximum
- Correlation is done with FFTs, since correlation in time is multiplication in frequency: roughly 17 billion operations become a few million

Step 5: Validation

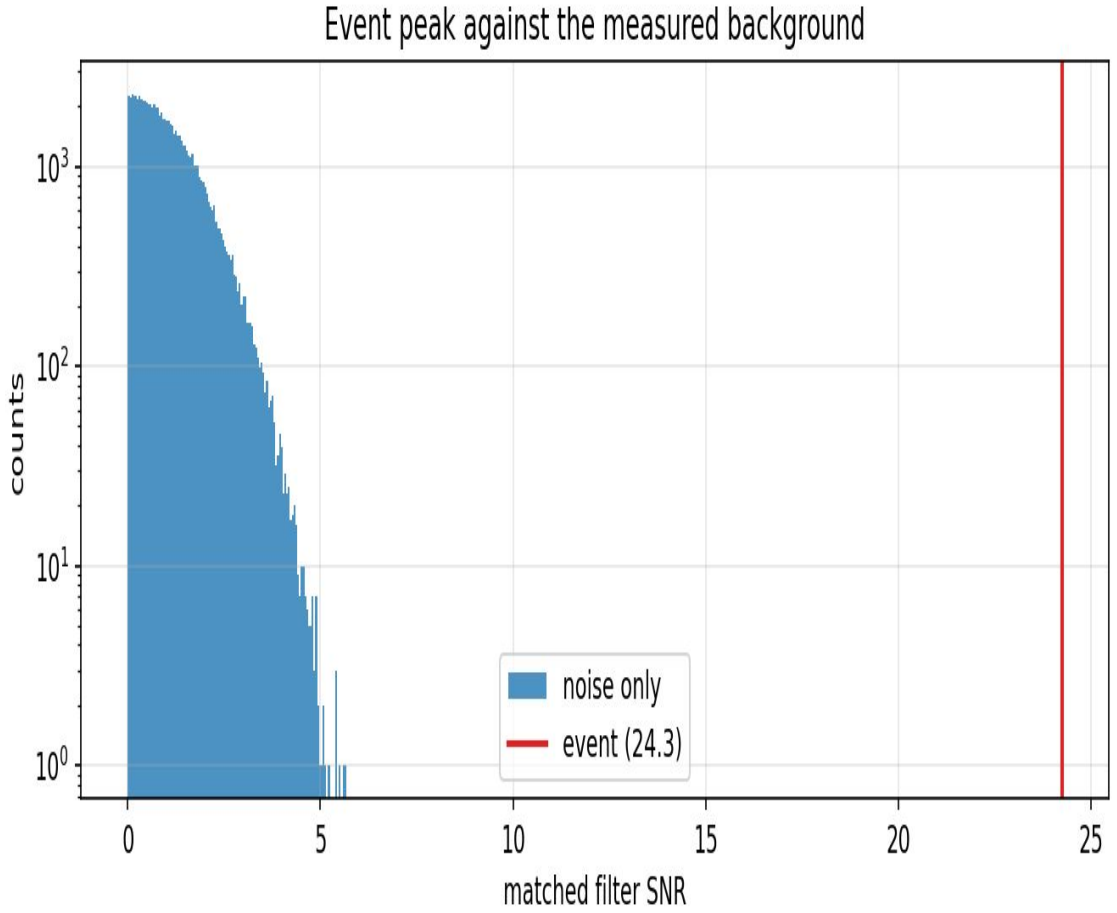
Two independent checks, neither of which assumes an answer

Check 1 · Detector coincidence

The two sites are 3000 km apart, therefore light takes 10 ms to cross between them. Measured arrival difference: 7.32 ms. A local instrument glitch would fail this test.

Check 2 · Measured background

I did not assume a threshold. I measured what noise alone produces in this same record: 81,920 samples, mean SNR 1.10, loudest noise-only peak 5.67. The event reached 24.25, a factor of 4.3 above the loudest noise.



What worked, What didn't work, Recommended changes

What worked

- Recovered a signal well below the noise floor in both detectors independently
- Arrival times agree to within the light travel time between sites
- Recovered the arrival time within 0.4 ms of published

What didn't work

- Whitened variance was 6x too high; traced to FFT edge effects, not the whitener
- Recovered SNR runs high: 24.25 for H1 against LIGO's published 20, and 17.92 for L1 against 13
- Cause: likely a single fixed template with no maximization over parameters, the PSD estimated on-source and a different integration band than the production pipeline

Recommended changes

- Estimate the PSD from off-source data only
- Use overlapping windowed segments instead of trimming
- Add a chi-squared test to reject glitches that score high without matching the chirp shape
- Run a template bank and maximize the SNR over the available templates

Data: LIGO Open Science Center (gwosc.org). Method: LIGO's published analysis chain; Abbott et al., PRL 116, 061102 (2016).